Specification for Approval

Feature

- water clear type
- 0402 top view
- superior weather-resistance
- high radiant intensity
- RoHS compliant

Appearance

Applications

- Backlighting in dashboard and switch
- Telecommunication: indicator and backlighting in telephone and fax
- Flat backlight for LCD ,switch and symbol
- Floppy disk drive
- Medical equipment
- Indication light on hand held products

Specification for Approval

Absolute Maximum Rating (Ta=25℃)

Item	Symbol	Value	Unit
DC forward current	I_F	20	mA
Pulse forward current*	I_{FP}	30	mA
Power dissipation	P_D	60	mW
Operating temperature	T_{opr}	-40~+85	℃
Storage temperature	Y_{stg}	-40~+80	℃
Reverse voltage	V_R	5	V
Sold soldering temperature	T_{sol}	260℃/3Sec	---

*Plus with Max 10ms,duty ratio max1/10

■ **Initial Electrical/Optical Characteristics** (Ta=25℃)

Item	Symbol	condition	Min	Type	Max	Unit
DC forward Voltage	V_F	I=20mA	1.8	2.0	2.3	V
DC reverse Current	I_R	V=5V	----	----	10	μA
Dome Wavelength	W_D	I=20mA	568	570	573	Nm
Spectrum Radiation Bandwidth	Δλ	I=20mA	----	20	----	nm
Luminous Intensity	Iv	I=20mA	55	65	75	mcd
50%Power Angle	2θ1/2	I=20mA	----	120	----	deg

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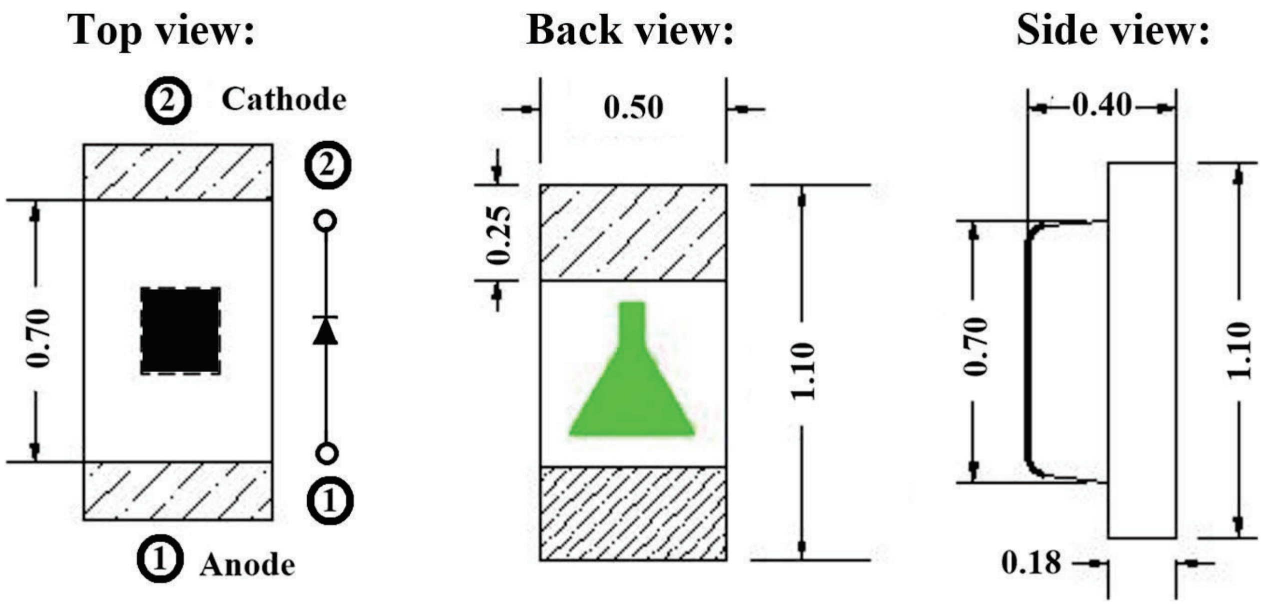
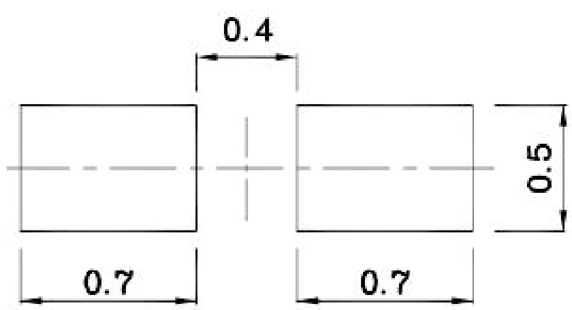
Package Schematics



Recommended Soldering Pattern



Notes:

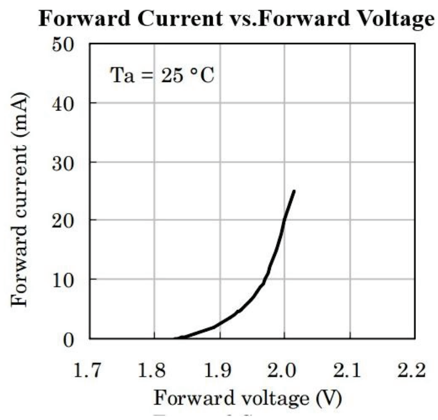
1. All dimensions are in millimeters (inches).

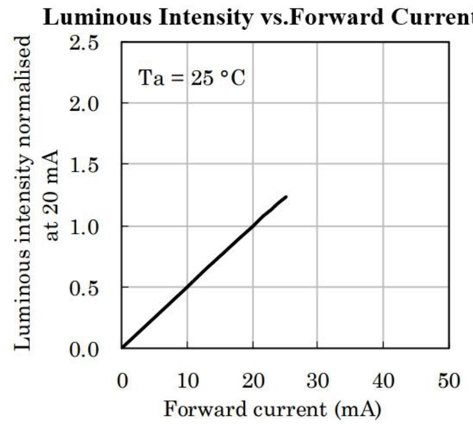
2.Tolerance is ±0.1(0.004") unless otherwise noted.

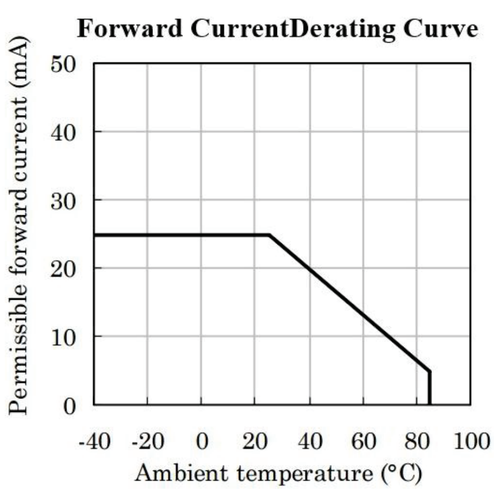
3. Specifications are subject to change without notice.

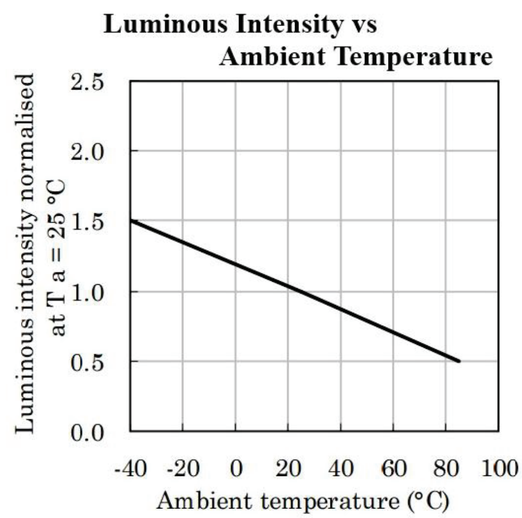
Specification for Approval

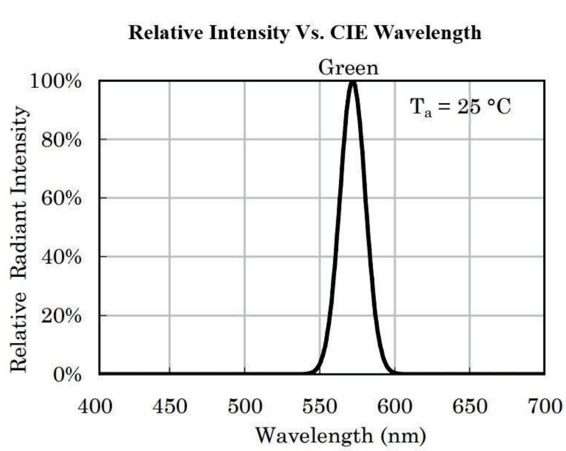
Typical Electro-Optical Characteristics Curves

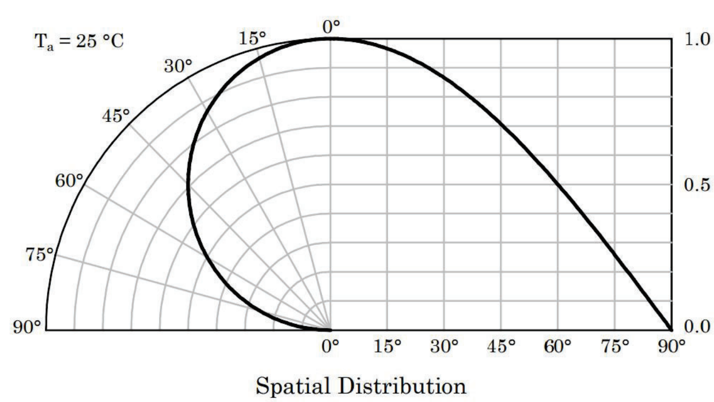












Specification for Approval

Reliabibity performance

Test items and result

Test classification	Test item	Test condionts	Test duration	Sample size	AC/RE
Life test	Room temperature DC operating life test	Ta=25±5ºC IF=20mA	1000hrs	30pcs	0/1
Environment test	Thermal shock Test	-10±5ºC←→+100±5ºC 5min 10sec 5min	50cysles	30pcs	0/1
	Temperature cycle test	-40±5ºC←→+85±5ºC 30min 5sec 30min	50cysles	30pcs	0/1
	High temperature & High humidity test	Ta=85±5ºC RH=85%±0.5%RH	1000hrs	30pcs	0/1
	High temperature storage	Ta =100±5 ºC	1000hrs	30pcs	0/1
	Low temperture storage	Ta =-55±5ºC	1000hrs	30pcs	0/1
Mechanical test	Resistance to soldering heat	Ta =230±5ºC	5sec	30pcs	0/1
	Lead integrity	Load 2.5N(0.25KGf) 0 ºC ∽ 90 ºC ∽ 0ºC	3times	30pcs	0/1

Specification for Approval

Reflow profile

Soldering condition

• Recommended soldering conditions

Reflow Soldering		Hand Soldering	
Pre-heat Pre-heat time Peak temperature Soldering time Condition	160～180℃ 120 seconds Max. 260℃ Max. 10 seconds Max. Refer to Temperature-profile	Temperature Soldering time	300℃ Max. 3 second Max. (one time only)

After reflow soldering rapid cooling should be avoided

Temperature-profile (Surface of circuit board)

Use the following conditions shown in the figure.

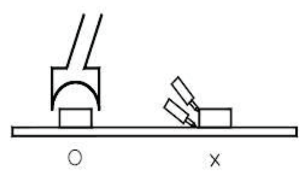
1. Reflow soldering should not be done more than two times
2. When soldering ,do not put stress on the LEDs during heating

Soldering iron

1. When hand soldering, keep the temperature of the iron under 300℃, and at that temperature keep the time under 3sec.
2. The hand soldering should be done only a time
3. The basic spec is ≤5 sec. when the temperature of 260℃, do not contact the resin when hand soldering

Rework

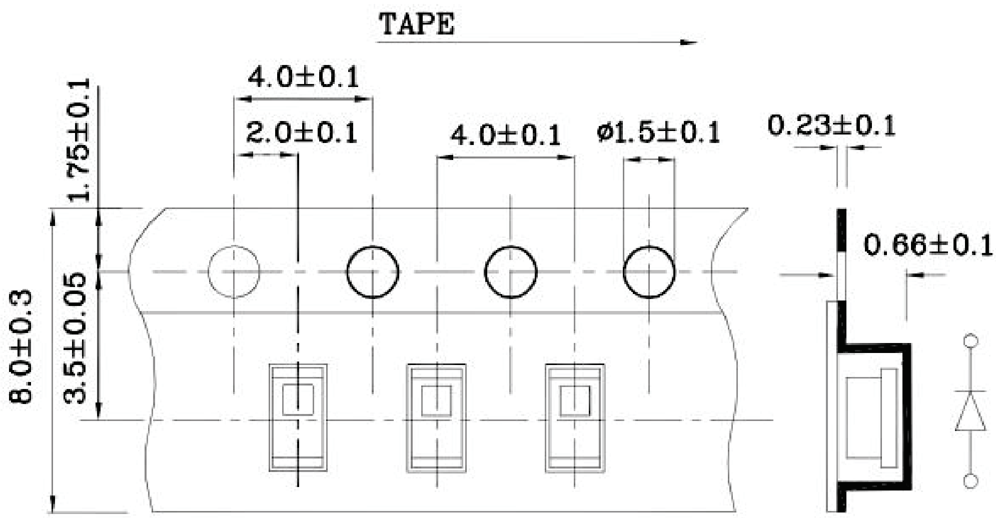
1. Customer must finish rework within 5 sec under 260℃
2. The head of iron can not touch the resin
3. Twin-head type is preferred.



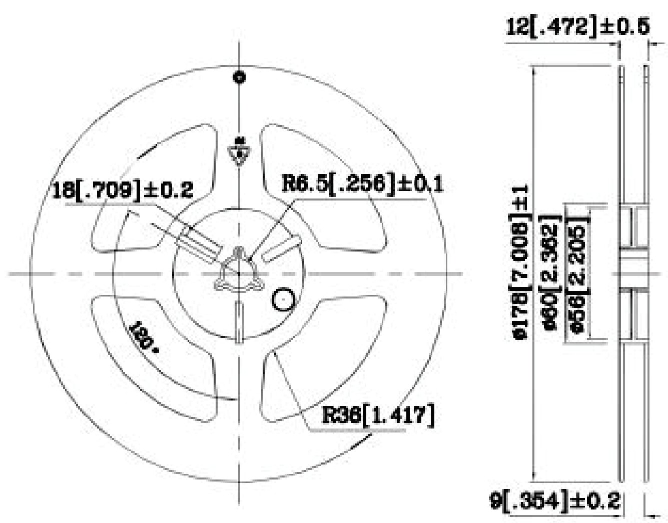
CAUTIONS

The encapsulated material of the LEDs is silicone . Therefore the LEDs have a soft surface on the top of package. The pressure to the top surface will be influence to the reliability of the LEDs. Precautions should be taken to avoid the strong pressure on the encapsulated part. So when using the picking up nozzle, the pressure on the silicone resin should be proper

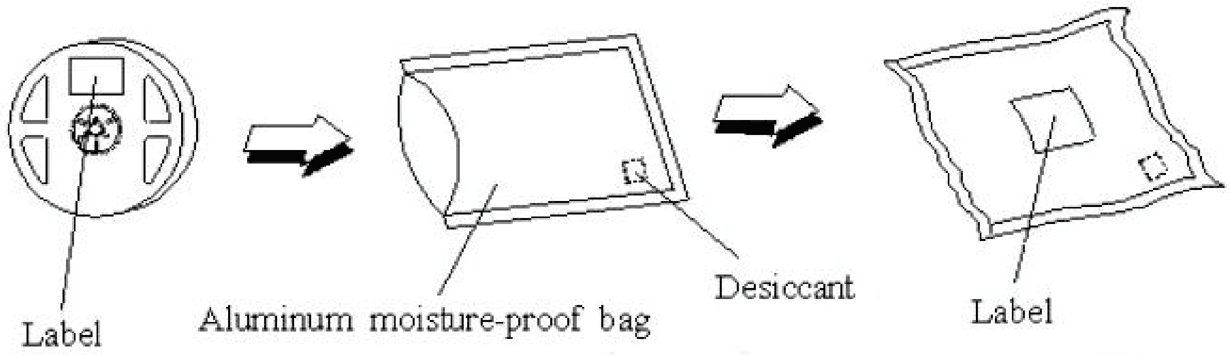
◆ Dimens ions for Tape



Reel Dimensions



◆ Dimensions for Reel



Notes:

1.All dimensions are in mm, tolerance is±2.0mm unless otherwise noted.

2.Specifications are subject to change without notice.